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Experiment 1 – Design Star and Snowflake Schema

Aim

To design **Two Star Schema** and **Snowflake Schema** for a Movie Ticket Booking system using a suitable fact table and dimension tables.

Objective

- To understand the concept of **Data Warehouse Schema**.
- To design a **Star Schema**.
- To design a **Snowflake Schema**.

Theory

⟶ What is Data Warehouse Schema ?

Data warehouse schema is the logical blueprint that shows how data is organized, connected, and used for analysis in a data warehouse . It defines the tables, keys, relationships, and rules that help BI tools and analysts query data consistently.

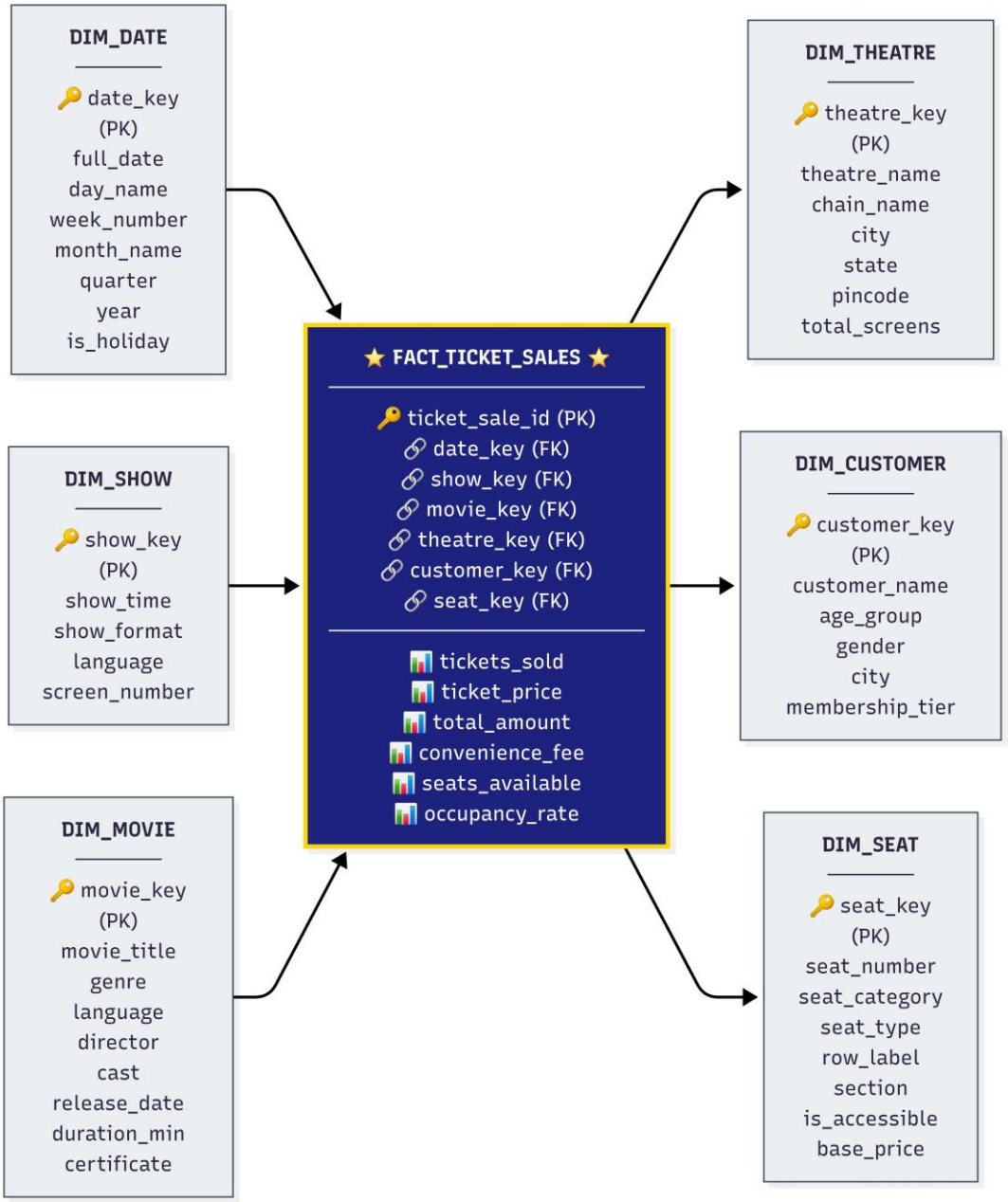
What it includes :

- Fact tables: store measurable business events, like sales or orders .
- Dimension tables: store descriptive details, like customer, product, or date .
- Relationships: connect facts and dimensions so analytical queries work correctly .

1. Star Schema

A star schema is a way of organizing data in a data warehouse for reporting and analytics. It is called a star schema because the structure looks like a star: a central fact table is connected to several surrounding dimension tables.

Structure of Star schema (Ticket Sales) :



Fact Tables :

Ticket_sales

Column	Description
ticket_sale_id	Primary Key
date_key	Foreign Key
show_key	Foreign Key
theatre_key	Foreign Key
customer_key	Foreign Key
seat_key	Foreign Key
Total_Amount	Total sales amount

Dimension Tables

Date_Dimension

Column	Description
Date_ID	Primary Key
Date	Date of transaction
Day	Day
Month	Month
Quarter	Quarter
Year	Year

Show_Dimension

Column	Description
Show_key	Primary Key
show_time	Foreign Key
show_format	Foreign Key
language	Foreign Key
screen_number	Foreign Key

Movie_Dimension

Column	Description
movie_key	Primary Key
movie_title	Foreign Key
genre	Foreign Key
language	Foreign Key
director	Foreign Key
cast	Foreign Key
release_date	Foreign Key
duration_min	Foreign Key
certificate	Foreign Key

Theatre_Dimension

Column	Description
theatre_key	Primary Key
theatre_name	Foreign Key
chain_name	Foreign Key
city	Foreign Key
state	Foreign Key
pincode	Foreign Key
total_screens	Foreign Key

Customer_Dimension

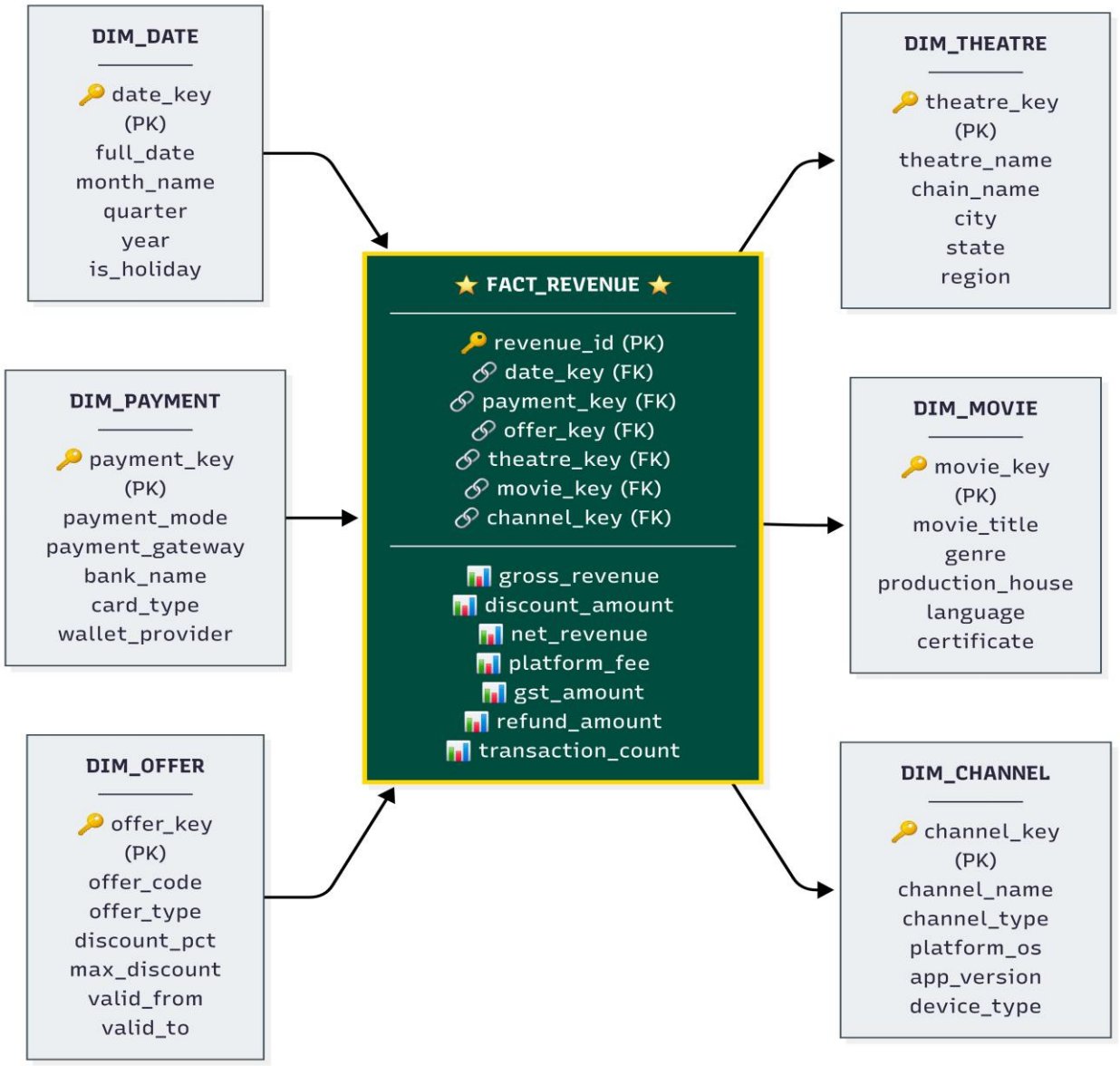
Column	Description
customer_key	Primary key
customer_name	Foreign Key
age_group	Foreign Key
gender	Foreign Key
city	Foreign Key
membership_tier	Foreign Key

Seat_Dimension

Column	Description
seat_key	Primary key
seat_number	Foreign Key
seat_category	Foreign Key
seat_type	Foreign Key
row_label	Foreign Key

section	Foreign Key
is_accessible	Foreign Key
base_price	Foreign Key

Structure of Star schema (Revenue)



Fact table:
Revenue :

Column	Description
revenue_id	Primary Key
date_key	Foreign Key
payment_key	Foreign Key
offer_key	Foreign Key
theatre_key	Foreign Key
movie_key	Foreign Key
channel_key	Foreign Key

Dimension tables :

Date_Dimension

Column	Description
date_key	Primary Key
full_date	Foreign Key
month_name	Foreign Key
quarter	Foreign Key
year	Foreign Key
is_holiday	Foreign Key

Payment_Dimension

Column	Description
payment_key	Primary Key
payment_mode	Foreign Key
payment_gateway	Foreign Key
bank_name	Foreign Key
card_type	Foreign Key
wallet_provider	Foreign Key

Offer_Dimension

Column	Description
offer_key	Primary Key
offer_code	Foreign Key
offer_type	Foreign Key
discount_pct	Foreign Key
max_discount	Foreign Key
valid_from	Foreign Key
valid_to	Foreign Key

Column	Description
theatre_key	Primary Key
theatre_name	Foreign Key
chain_name	Foreign Key
city	Foreign Key
state	Foreign Key
region	Foreign Key

Movie_Dimension

Column	Description
movie_key	Primary key
movie_title	Foreign Key
genre	Foreign Key
production_house	Foreign Key
language	Foreign Key
certificate	Foreign Key

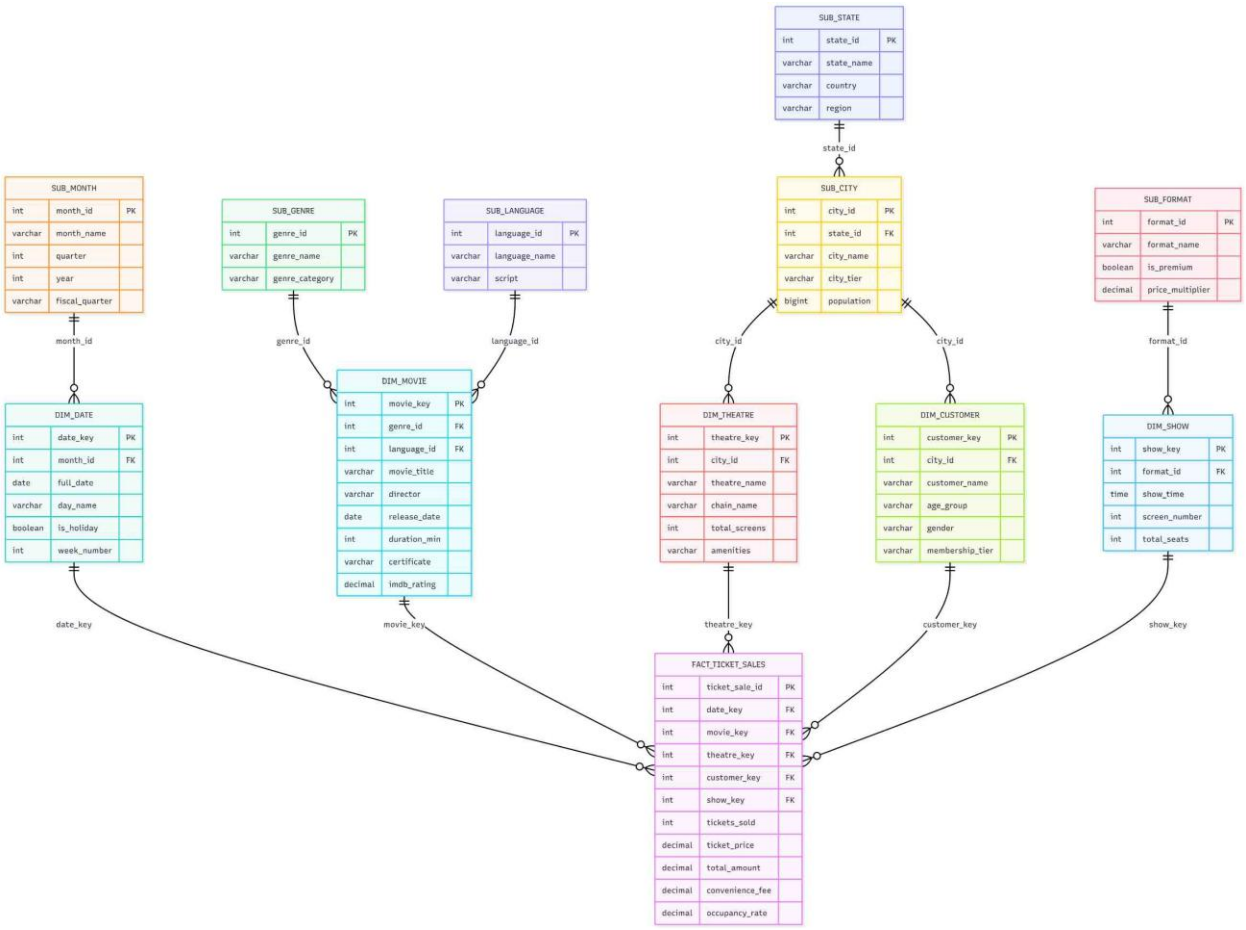
Channel_Dimension

Column	Description
channel_key	Primary Key
channel_name	Foreign Key
channel_type	Foreign Key
platform_os	Foreign Key
app_version	Foreign Key
device_type	Foreign Key

2 . Snowflake Schema

A Snowflake Schema is a data warehouse schema design where dimension tables are normalized into smaller, separate sub-dimension tables. It extends the Star Schema structure to reduce data redundancy and save storage space by organizing attributes into hierarchical relationships.

Structure:



Main Difference :

Star schema uses a central fact table directly linked to fully denormalized

dimension tables, forming a star-like shape. All dimensional data is stored in a single table, which means fewer joins and faster query performance, but at the cost of data redundancy and higher storage usage.

Snowflake schema is an extension of the star schema where dimension tables are further normalized into sub-dimension tables. For example, instead of storing city, state, and country in a single Dim_Location table, each gets its own table linked hierarchically. This reduces redundancy and saves storage, but requires more joins, making queries more complex and relatively slower.

Result :

Thus, the Star Schema and Snowflake Schema for the Movie Ticket booking system were successfully designed by identifying the Sales & Revenue Fact Table and corresponding Dimension Tables.

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GitHub Repository Link:

https://github.com/Sentry2311/DMBI_EXPERIMENTS/tree/main/Experiment_01_Star_and_Snowflake_Schema

(Contains complete documentation, dataset, and program code for Experiment 1: Star and Snowflake Schema)